

The Mathematics of the Thesis
Identifiability, Smile Transport and Finite-Sample Model-Risk Certification
for the JIBAR-to-ZARONIA Derivatives Book

Revised after council review, and re-cut to a three-year programme

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Abstract

This document states every mathematical result the thesis promises, at proposal precision: hypotheses, statement, proof route, failure mode. It is the second version. The first version was put before three reviewers acting as a mathematical statistician, a mathematical finance theorist and a doctoral examiner; their corrections are incorporated and marked in the margin. Two corrections were structural. First, the direction of the identifiability argument in the finance half was reversed: the unidentified correlations contaminate the observed JIBAR smile, not the target compounded-rate smile. Second, the statistics half as first drafted assembled known results and would not carry a mathematical-statistics doctorate; a new theorem on optimal block length with a matching minimax lower bound (Theorem 9) is now the spine. The final section re-cuts the programme to thirty-six months.

Summary of council corrections

Result	Change forced by review
Lemma 1	Same- g scaling is valid without Dambis–Dubins–Schwarz; linear g and the factor $\Delta/3$ are assumptions, not theorems. Non-decaying vol-of-vol added as Proposition 2.
Theorem 3	JIBAR caplet is priced under $\mathbb{Q}^{T+\Delta}$, not $\mathbb{Q}^T$; the in-advance/in-arrears convexity term does not exist and is deleted. Basis correlations enter the JIBAR smile, not the compounded caplet. Two unidentified correlations, not one. Expansion parameter redefined.
Theorem 4	Becomes a corollary of Theorem 3; difference is at the money, not wing-only; rank condition is trivial. Demoted from difficulty 4 to 2–3.
Proposition 5	“Static arbitrage” replaced by “model inconsistency”; exact only for normal SABR with equal volatilities; otherwise leading order in variance. Demoted to proposition, appendix in the three-year plan.
Theorem 6	Coupling cost $(n+1)\beta(b)$, not $2n\beta(b)$; deployment block must be gap-separated; conditional form uses the Beta law of conformal coverage, not DKW; drift penalty is Kolmogorov distance $d_K \leq d_{TV}$, applied after coupling.
Proposition 7	The first version was false (counterexample: point mass versus narrow uniform). Correct bound is $\sqrt{2LW_1}$; plug-in rate $n^{-1/4}$; the dependent-data version of the empirical W_1 bound is currently missing.
Theorem 8	$\delta_{+\infty}$ is not a measure on $\mathbb{R}$; construction replaced; the sharp constant is 1, not $1 - \alpha$; the argument is adversarial, not Le Cam.
First-draft bootstrap theorem	Dropped as a theorem. Empirical comparison with block bootstrap retained.
Theorem 9	New. Optimal block length and minimax rate over a mixing class. This is the doctoral theorem.

1 Setting and notation

Fix a filtered probability space carrying a Brownian motion (W, Z) with $d\langle W, Z \rangle_t = \rho dt$, and a third Brownian motion W^B with $d\langle W^B, W \rangle_t = \rho_{BF} dt$, $d\langle W^B, Z \rangle_t = \rho_{B\alpha} dt$. Let $P(t, U)$ denote the zero-coupon bond price and $\mathbb{Q}^U$ the U -forward measure.

Compounded rate. For an accrual window $[T, T + \Delta]$ with daily fixings r_i and day-count fractions δ_i ,

$$R(T, T + \Delta) = \frac{1}{\Delta} \left(\prod_i (1 + r_i \delta_i) - 1 \right) \approx \frac{1}{\Delta} \left(\exp \left(\int_T^{T+\Delta} r_u du \right) - 1 \right). \quad (1)$$

The continuous approximation is used throughout; it makes $1 + \Delta F_t = P(t, T)/P(t, T + \Delta)$ exact and differs from the daily product at second order in $r_i \delta_i$, below a basis point. [Council correction: state once, keep continuous form]

Forwards. The forward compounded rate $F_t := \mathbb{E}^{\mathbb{Q}^{T+\Delta}}[R(T, T+\Delta) \mid \mathcal{F}_t]$ is a $\mathbb{Q}^{T+\Delta}$ -martingale. The JIBAR forward L_t for the same window is fixed at T and *paid at* $T + \Delta$; it is therefore also a $\mathbb{Q}^{T+\Delta}$ -martingale, and so is the basis $B_t := L_t - F_t - s$, where s is the fixed credit adjustment spread. In particular $\mathbb{E}^{\mathbb{Q}^{T+\Delta}}[B_T] = B_0 = L_0 - F_0 - s$ is read off the two curves. [Council correction: both caplets live under $\mathbb{Q}^{T+\Delta}$; no in-advance/in-arrears convexity term]

Decay and clock. Let g be deterministic and bounded with $g(t) = 1$ for $t \leq T$ and $g(t) \downarrow 0$ on $(T, T + \Delta]$, and set $\tau(t) := \int_0^t g(u)^2 du$.

Assumption 1 (Linear decay). $g(t) = (T + \Delta - t)/\Delta$ on $(T, T + \Delta]$, so that $\tau(T + \Delta) = T + \Delta/3$.

[Council correction: linear g is exact only when the instantaneous forward-rate volatility is flat in maturity; generically $\tau(T + \Delta) = T + c_g \Delta$ with $c_g \in (0, 1)$. Every result below inherits this assumption.]

Dynamics. Under $\mathbb{Q}^{T+\Delta}$,

$$dF_t = g(t) \alpha_t F_t^\beta dW_t, \quad d\alpha_t = g(t) \nu \alpha_t dZ_t, \quad dB_t = \eta dW_t^B. \quad (2)$$

The dimensionless basis scale is $\varepsilon := \eta/(\alpha_0 F_0^\beta)$. [Council correction: $\eta\sqrt{T}$ is a rate, not a small parameter]

Scores. For a fixed valuation-and-hedging policy π , the hedging error over a block of b consecutive rebalancing periods is the *score* S . Calibration scores have law P , deployment scores law Q . F_P, F_Q are their distribution functions, $d_K(P, Q) = \sup_x |F_P(x) - F_Q(x)| \leq d_{TV}(P, Q)$ the Kolmogorov distance.

2 Finance results

Lemma 1 (Time-changed SABR; Willems 2020, Lyashenko–Mercurio 2019). *Under (2) with the same g on both diffusion coefficients, $(F_t, \alpha_t) = (\tilde{F}_{\tau(t)}, \tilde{\alpha}_{\tau(t)})$ where $(\tilde{F}, \tilde{\alpha})$ is constant-parameter SABR($\alpha_0, \beta, \rho, \nu$). Hence*

$$P(0, T + \Delta) \mathbb{E}^{\mathbb{Q}^{T+\Delta}}[(F_{T+\Delta} - K')^+] = P(0, T + \Delta) C^{\text{SABR}}(F_0, K', \tau(T + \Delta); \alpha_0, \beta, \rho, \nu), \quad (3)$$

with the Hagan approximation carrying its usual $O(\alpha^2 \tau)$ error.

Proof. Because τ is deterministic, $(\int_0^t g dW, \int_0^t g dZ)$ is a correlated Brownian pair in the clock τ ; substitute. [Council correction: no Dambis–Dubins–Schwarz needed; the same- g condition is what makes the substitution exact] $\square$

Proposition 2 (Non-decaying vol-of-vol; target). *If instead $d\alpha_t = \nu \alpha_t dZ_t$ (vol-of-vol does not run off, the physically defensible choice), then in the clock τ the vol-of-vol is $\nu/g(\tau^{-1}(u))$ and no closed form exists. The caplet is approximated by Hagan–Lesniewski–Woodward effective parameters $(\rho_{\text{eff}}, \nu_{\text{eff}})$; relative to Lemma 1 the total vol-of-vol variance is $\nu^2(T + \Delta)$ rather than $\nu^2(T + \Delta/3)$, so Lemma 1 understates smile curvature by $\frac{2}{3}\nu^2\Delta$ (about 17% for $T = 1y$, $\Delta = 3m$).*

Remark. Neither result is a contribution. Lemma 1 is cited; Proposition 2 is a two-page derivation in Chapter 2 and fixes which variant the empirical chapters use.

Theorem 3 (Identified and unidentified components of the smile map; target). *Let (2) hold with Assumption 1, fix β , and suppose $\nu \neq 0$. Denote by $\sigma_L^N(K, T)$ the observed JIBAR normal implied volatility and by $\sigma_F^N(K', T + \Delta)$ the compounded-rate normal implied volatility at the fallback strike $K' = K - s$. Then*

$$\sigma_F^N(K', T + \Delta) = \Phi_0(K'; \sigma_L^N(\cdot, T), s, B_0, \Delta) - \rho_{BF} \eta \Phi_1(K') - \rho_{B\alpha} \eta \nu \Phi_2(K') + O(\varepsilon^2), \quad (4)$$

where

1. Φ_0 is the SABR smile at clock $T + \Delta/3$ whose triple (α_0, ρ, ν) is fitted to $\sigma_L^N(K' + s + B_0, T)$ at clock T : a strike shift by $s + B_0$ and the time change of Lemma 1;
2. $\Phi_1(K') = 1 + O(K' - F_0)$: a parallel level shift;
3. $\Phi_2(K') \propto (K' - F_0)$: a skew tilt.

Consequently: the curvature parameter ν and the map Φ_0 are identified from the observable set $\mathcal{S} = \{\sigma_L^N(\cdot, T)\} \cup \{B_0\} \cup \{s\}$; the pair $(\eta\rho_{BF}, \eta\rho_{B\alpha})$ is not identified at $O(\varepsilon)$, and $(\eta, \rho_{BF}, \rho_{B\alpha})$ is not identified at $O(\varepsilon^2)$. In particular the successor at-the-money volatility is identified only within an interval of half-width η .

[Council correction: the first draft had the unidentified correlation entering the compounded caplet; it is the other way round. The compounded caplet depends only on $(\alpha_0, \beta, \rho, \nu)$. The JIBAR smile is a contaminated view of that triple, because $L = F + B + s$ and the basis noise shifts its level (through ρ_{BF}) and tilts its skew (through $\rho_{B\alpha}\nu$). The convexity term was deleted because both instruments pay at $T + \Delta$.]

Proof route. Under $\mathbb{Q}^{T+\Delta}$ write $L = F + B + s$ and expand the JIBAR caplet in ε : the level of σ_L^N is σ_F^N -at-clock- T plus $\rho_{BF}\eta$ at first order (variance of a sum), the skew acquires $\rho_{B\alpha}\eta\nu$ from $\text{Cov}(B, \alpha)$, and the curvature is unaffected at $O(\varepsilon)$. Invert: fitting SABR to σ_L^N returns $(\alpha_0 - \rho_{BF}\eta + \dots, \rho + c\rho_{B\alpha}\eta\nu/\alpha_0 + \dots, \nu)$. Push the fitted triple through the time change to obtain Φ_0 ; the residual terms are Φ_1, Φ_2 .

Failure mode. If the basis has mean reversion comparable to $1/T$, the $O(\varepsilon)$ coefficients acquire T -dependence and the interval half-width becomes $\eta h(\kappa T)$ for an explicit h . The statement survives with a modified constant.

Theorem 4 (Two models; corollary of Theorem 3). *For any (v_1, v_2) with $|v_i| \leq \eta$ there exist models $\mathcal{M}_\pm$ with $(\eta\rho_{BF}, \eta\rho_{B\alpha}) = \pm(v_1, v_2)$ that match every JIBAR caplet to $O(\varepsilon^2)$ and match B_0 exactly, while their compounded-rate at-the-money volatilities differ by $2v_1$ and their skews by $2v_2\nu$, at every strike.*

Proof. Flip the sign of $(\rho_{BF}, \rho_{B\alpha})$ and re-solve (α_0, ρ) so that the first-order level and skew of σ_L^N are unchanged. The Jacobian $\partial(\text{level, skew})/\partial(\alpha_0, \rho)$ is nonsingular for $\nu \neq 0$, so the re-solve exists. Read the differences off (4). [Council correction: rank is immediate, not a research question; difference is at the money, not wing-only] $\square$

Remark (Parameter count). With m expiries and β fixed: observables are $3m$ smile features (level, skew, curvature per expiry; additional strikes test SABR shape and carry η only at $O(\varepsilon^2)$) plus m basis points, which pin $B_{0,i}$ and hence strike shifts but no smile parameters. Unknowns are $3m$ SABR parameters plus $(\eta, \rho_{BF}, \rho_{B\alpha})$ if shared across expiries. Deficit: 3 (two at $O(\varepsilon)$). If the basis parameters are per-expiry, the deficit is $3m$. Historical basis data estimate the nuisance under the physical measure, not the pricing measure; that estimate is the ambiguity set that Theorem 6 takes as input.

Proposition 5 (Tenor consistency; [target](#), appendix in the three-year plan). *For $t \leq T_1 < T_2 < T_3$, $(1 + \Delta_{13}F_t^{13}) = (1 + \Delta_{12}F_t^{12})(1 + \Delta_{23}F_t^{23})$ exactly (ratios of zero-coupon bonds).*

- (a) *If $\beta = 0$, g linear and (W, Z) common, the induced R_{13} is time-changed normal SABR with linear decay if and only if $\alpha_{12} = \alpha_{23}$; then $\alpha_{13} = \alpha_{12}$, (ρ, ν) are inherited, and*

$$\Delta_{13}^2\tau_{13} = \Delta_{12}^2\tau_{12} + \Delta_{23}^2\tau_{23} + 2\Delta_{12}\Delta_{23}c, \quad c = T_1 + \Delta_{12}/2,$$

exactly; the clocks compose.

- (b) *Otherwise the displayed relation is the variance-matched projection, valid to leading order in $\alpha^2\tau$, and the constraints tying (ρ, ν) across tenors are approximate.*
- (c) *Violation of (a)–(b) means no model in the class calibrates jointly to the 3M and 6M smiles. It is not a static arbitrage: $(R_{13} - K)^+$ is not spanned by 3M caplets. [Council correction: a model-free statement would need Hobson–Laurence–Wang-type bounds for an option on a product with given marginals, which is a different theorem and is not claimed]*

3 Statistics results

Assumption 2 (Mixing and separation). The per-period hedging-error process of the fixed policy π is β -mixing with coefficients $\beta(k)$ (non-stationary definition: supremum over time origins) and stationary on the calibration window. The fitting sample $\mathcal{D}_{\text{fit}}$, the n calibration blocks of length b , and the deployment block are pairwise separated by gaps of length at least b . [Council correction: the deployment gap is required; without it S_0 is not controlled at all]

Let $k := \lceil (1 - \alpha)(n + 1) \rceil$, $\hat{q} := S_{(k)}$ the k -th order statistic of the calibration scores, and $b_{n,k}(\delta) := \text{Beta}^{-1}(\delta; k, n + 1 - k)$ the δ -quantile of the $\text{Beta}(k, n + 1 - k)$ law.

Theorem 6 (Coverage under mixing and drift; **target**). *Under Assumption 2, with calibration scores $S_1, \dots, S_n \sim P$ marginally and deployment score $S_0 \sim Q$:*

- (i) (Marginal.) $\mathbb{P}(S_0 \leq \hat{q}) \geq 1 - \alpha - d_K(P, Q) - (n + 1)\beta(b)$.
- (ii) (Conditional.) For any $\delta, \delta' \in (0, 1)$, with probability at least $1 - \delta - n\beta(b)$ over $(\mathcal{D}_{\text{fit}}, S_{1:n})$,

$$\mathbb{P}(S_0 \leq \hat{q} \mid \mathcal{D}_{\text{fit}}, S_{1:n}) \geq b_{n,k}(\delta) - d_K(P, Q) - \beta(b)/\delta',$$

the last term holding on an event of probability at least $1 - \delta'$. Moreover $b_{n,k}(\delta) \geq 1 - \alpha - \sqrt{\log(1/\delta)/(2n)}$, and $b_{n,k}(\delta) \approx 1 - \alpha - \sqrt{2\alpha(1 - \alpha)\log(1/\delta)/n}$, which is about 2.3 times tighter than the DKW form at $\alpha = 0.05$.

No assumption is made that the model underlying π is correct.

[Council correction: coupling cost is $(n + 1)\beta(b)$ (Yu 1994, Lemma 4.1: $(m - 1)\beta(b)$ for m blocks, plus independence from $\mathcal{D}_{\text{fit}}$ and the deployment block); the conditional statement cannot be obtained from a joint total-variation bound and uses the β -mixing definition with Markov's inequality; the conformal coverage is $\text{Beta}(k, n + 1 - k)$ -distributed (Vovk 2012), so DKW is a corollary, not the tool; the drift penalty is Kolmogorov distance because only half-lines are evaluated, applied to the fixed set $(-\infty, \hat{q}^*]$ after coupling has made S_0^* independent of $\hat{q}^*$.]

Proof route. (i) Sequential Berbee coupling replaces the $n + 1$ separated blocks and $\mathcal{D}_{\text{fit}}$ by independent copies $S_{0:n}^*$ on an event of probability $\geq 1 - (n + 1)\beta(b)$. (ii) On that event, conditional on π , the calibration copies are i.i.d. P and $F_P(\hat{q}^*) \sim \text{Beta}(k, n + 1 - k)$, so $\mathbb{P}(S_0^* \leq \hat{q}^*) \geq 1 - \alpha$ if $S_0^* \sim P$. (iii) For $S_0^* \sim Q$ and the fixed half-line $(-\infty, \hat{q}^*]$, $|Q(S \leq \hat{q}^*) - P(S \leq \hat{q}^*)| \leq d_K(P, Q)$. Sum.

Remark. $\beta(b)$ is not estimable from data (Adams–Nobel-type impossibility). The certificate is conditional on an *assumed* mixing class $\{\beta(k) \leq Ck^{-r}\}$, and the thesis says so. [Council correction: added]

Proposition 7 (Drift penalty bounded by W_1 ; **target**). *If Q has a density bounded by L (no assumption on P), then for all x , $F_P(x) - F_Q(x) \leq \sqrt{2LW_1(P, Q)}$. If both have densities bounded by L , $d_K(P, Q) \leq \sqrt{LW_1(P, Q)}$. Also $F_P(x) - F_Q(x) \leq LW_\infty(P, Q)$.*

Proof. Let $D = F_P - F_Q$. If $D(x_0) = d > 0$ then, since F_P is non-decreasing and F_Q is L -Lipschitz, $D(x) \geq d - L(x - x_0)$ on $[x_0, x_0 + d/L]$, whence $W_1(P, Q) = \int |D| \geq d^2/(2L)$. The W_∞ bound uses the monotone coupling: $F_P(x) \leq F_Q(x + W_\infty) \leq F_Q(x) + LW_\infty$. $\square$

[Council correction: the first version claimed $|F_P - F_Q| \leq LW_1$, which is false: $P = \delta_0$, $Q = U[0, 1/L]$ has $W_1 = 1/(2L)$ and CDF gap 1. With the square-root form the plug-in penalty decays at $n^{-1/4}$, not $n^{-1/2}$. The i.i.d. rate $\mathbb{E}W_1(P_n, P) \leq n^{-1/2}J_1(P)$ is Bobkov–Ledoux (2019) in $d = 1$; deployment scores are dependent, so a β -mixing version (Dedecker–Merlevède type) is needed and is part of Theorem 9(iii). L is an assumption on the unobserved Q and is stated as such.]

Theorem 8 (Sharpness of the drift penalty). *Let T be any threshold procedure independent of the deployment score, and $C_T(R) := \mathbb{P}_{S \sim R}(S \leq T)$. For every P with continuous distribution function and every $\varepsilon \in (0, 1)$:*

- (a) (Universal.) $\inf_{Q: d_{\text{TV}}(P, Q) \leq \varepsilon} C_T(Q) \geq C_T(P) - \varepsilon$.
- (b) (Attained.) *Let Q_M be P with its bottom- ε mass on $(-\infty, q_\varepsilon(P)]$ moved to the point M . Then $d_{\text{TV}}(P, Q_M) = d_K(P, Q_M) = \varepsilon$ and*

$$C_T(Q_M) \leq C_T(P) - \varepsilon + \varepsilon \mathbb{P}(F_P(T) < \varepsilon) + \varepsilon \mathbb{P}(T \geq M).$$

For split conformal, $\mathbb{P}(F_P(T) < \varepsilon) = \mathbb{P}(\text{Beta}(k, n+1-k) < \varepsilon)$, exponentially small for $\varepsilon < 1 - \alpha$. Hence the constant 1 on d_K in Theorem 6 is sharp.

Proof. (a) is the definition of total variation applied to the event $\{S \leq T\}$ conditional on T . (b) Under Q_M , mass ε that P placed below every plausible threshold now sits at M ; if $T < M$ and $F_P(T) \geq \varepsilon$, coverage falls by exactly ε . The two correction terms cover the complementary events. $\square$

[Council correction: $\delta_{+\infty}$ is not a probability measure on $\mathbb{R}$. The far-mass construction $(1 - \varepsilon)P + \varepsilon\delta_M$ gives only $(1 - \varepsilon)C_T(P)$, which is weaker and sharp only for degenerate procedures. Moving mass from the bottom of P to a far point is what attains the constant 1. Since Q is never observed, this is a direct adversarial construction, not Le Cam’s method; Le Cam enters only in Theorem 9(ii) where deployment scores are partially observed.]

Theorem 9 (Optimal block length and minimax rate; **target**, the doctoral theorem). *Fix the mixing class $\mathcal{B}_r := \{\beta(k) \leq Ck^{-r}\}$, $r > 1$, and a calibration record of N periods. Any blocking scheme with block length b and gaps b yields $n \asymp N/(2b)$ blocks, and by Theorem 6(i) its coverage deficit (excluding drift) is*

$$\mathcal{E}(b, N) \asymp \sqrt{b/N} + Nb^{-(r+1)}.$$

- (i) (Upper bound.) *The minimiser is $b^* \asymp N^{3/(2r+3)}$, giving $\mathcal{E}(b^*, N) \asymp N^{-r/(2r+3)}$.*
- (ii) (Lower bound.) *There exist processes in $\mathcal{B}_r$ (hidden-regime constructions) for which no threshold procedure based on N calibration periods achieves deficit smaller than $cN^{-r/(2r+3)}$. Hence the rate is minimax over $\mathcal{B}_r$. When m deployment scores are additionally observed, the minimax certifiable deficit over $\{Q : d_K(P, Q) \leq \varepsilon\}$ is obtained adaptively in unknown ε by Le Cam’s two-point method on the observed deployment sample.*
- (iii) (Dependent empirical W_1 .) *For β -mixing scores with $\beta \in \mathcal{B}_r$, $\mathbb{E} W_1(Q_m, Q) \leq c_r m^{-1/2} J_1(Q)$ with an explicit c_r , closing the gap left in Proposition 7.*

[Council correction: this theorem did not exist in the first draft. The statistician’s verdict on the first draft was that Theorems 6 and 8 together are Oliveira et al. (2024) glued to a one-line Kolmogorov shift bound plus a paragraph of folklore, with application-level novelty only. Theorem 9 is the result that makes the statistics half doctoral: a rate, an optimal design, and a matching lower bound over a natural class.]

Proof route. (i) Balance $\sqrt{b/N} = Nb^{-(r+1)}$. (ii) Construct a two-regime process in $\mathcal{B}_r$ whose regimes are indistinguishable from N periods when the regime persistence matches b^* ; standard Assouad or two-point reduction on the regime label. (iii) Adapt the Bobkov–Ledoux $d = 1$ argument to blocked sums via the same Berbee coupling used in Theorem 6.

Failure mode. The lower bound in (ii) may hold only up to a logarithmic factor, or over a slightly smaller class. Either outcome is publishable. The upper bound (i) is elementary and cannot fail.

Remark (On the first-draft bootstrap theorem). The comparison with the stationary block bootstrap is no longer stated as a theorem. Bootstrap coverage rates for quantiles under mixing require Edgeworth expansions with unknown constants, and “the only method with a provable finite-sample level” is a comparison of assumptions, not a result. The comparison is retained as an empirical section on USD and ZAR data. [Council correction: dropped]

4 Difficulty after correction

Result	Kind	Deepest tool	Difficulty (1–5)	Carries the doctorate
Lemma 1	cited	deterministic time change	1	no
Prop. 2	derivation	HLW effective parameters	2	no
Thm 3	expansion, error bound	forward measure, SABR inversion	3	finance paper
Thm 4	corollary	nonsingular Jacobian	2	no
Prop. 5	algebra	variance matching	2–3	appendix
Thm 6	theorem	Berbee coupling, Beta law, d_K	3	supports
Prop. 7	proposition	Lipschitz CDF argument	2	supports
Thm 8	theorem	adversarial construction	2	supports
Thm 9	theorem	minimax lower bound, dependent empirical W_1	4	yes: the spine

Not needed anywhere: Malliavin calculus, rough paths, G-expectation, optimal transport, reflected BSDEs, deep-learning theory.

5 The three-year programme

Principles. The statistics spine is proved first, in the candidate’s home field, while finance prerequisites are learned in parallel: Theorem 6 needs only a fixed policy, and the market convention (strike shift, volatility scaling) supplies one without Theorem 3. The USD LIBOR-to-SOFR transition is the empirical spine, because the target smile became observable there and every input is on the university terminal; the South African book is a coda, because by month 30 it will have at most twenty-four monthly hedging-error observations and its option surfaces need a sponsor who may not sign.

Cuts. Proposition 5 to an appendix, stated with the exact case (a) proved and (b) as a conjecture. The first-draft bootstrap theorem to an empirical section. The standalone factual-record note folded into Chapter 1 at three thousand words. The standalone time-changed-SABR note becomes Chapter 2 preliminaries. The South African chapter shrinks to five thousand words.

Minimal passing thesis. Theorems 6, 8 and 9 proved; the USD certificate test with honest coverage; Theorem 3 with the $O(\varepsilon)$ identified part and Theorem 4 demonstrated numerically on USD; the South African coda on public data if no sponsor. This passes because the spine is proved, sharp, rate-optimal, and tested where the truth was observable.

Months	Activity	Milestone
1–4	Stochastic calculus and forward measures; Barber et al. (2023), Oliveira et al. (2024), Vovk (2012) line by line	Reading report
3–6	Theorem 8 proved; Theorem 6 route drafted with the Beta form	Thm 8 done (m5)
5–9	USD multi-curve bootstrap and time-changed SABR caplet code; Lemma 1, Prop. 2	Working code
7–10	Theorem 6 full proof; Proposition 7	Thm 6 done (m10)
9–11	Proposal defence: Chapters 1–2 drafts, Theorems 6, 8 proved, Theorem 9 stated	Faculty approval (m11)
10–12	USD data assembled; ZAR sponsor decision	Data agreement or public-data fallback (m12)
10–18	Theorem 9: upper bound (m12), lower-bound construction (m16), dependent W_1 (m18)	Thm 9 done (m18)
12–17	USD certificate test with optimal blocking; bootstrap and traffic-light comparison	Chapter 6 part one
14–20	Theorem 3 expansion; Theorem 4 construction, with finance co-supervisor	Chapter 4 draft (m20)
18–22	Paper 1: Theorems 6, 8, 9, Prop. 7, USD test	Submitted, <i>EJS</i> or <i>Bernoulli</i> (m22)
20–24	USD identifiability validation: predicted SOFR smile and $\pm\eta$ interval against the observed smile	Chapter 6 complete (m24)
24–28	Paper 2: Theorems 3, 4, USD identifiability	Submitted, <i>Quantitative Finance</i> (m28)
26–30	ZAR coda: JIBAR surface to Dec 2026, converted-book P&L to m30, reserve decomposition	Chapter 7 draft (m30)
30–34	Full write-up; Chapters 1, 8, appendices; internal review	Complete draft (m34)
34–36	Corrections, submission	Thesis submitted (m36)

Chapters and words. 1 Introduction and the transition (5k); 2 Preliminaries (8k); 3 Certification (10k); 4 Sharpness and optimal blocking (6k); 5 Identifiability (10k); 6 USD validation (10k); 7 South African coda (5k); 8 Conclusions (3k); appendices. About 57 thousand words.

Papers. Two, not three. Paper 1 is the doctoral paper. The South African material becomes a practitioner note to the Market Practitioners Group.

Probability of submission. Examiner’s estimate: about 60% by month 36; about 85% by month 42 with the customary extension. The plan passes without ZAR data and without Theorem 3 being clean. It does not pass without Theorem 6 by month 10–12, Theorem 9 by month 18, and the USD test by month 17.

Ranking under three years. Three years widens this topic’s lead over the alternatives the council reviewed. Conformal deep hedging on index options needs a year of training infrastructure before any theorem. Certified superhedging under volatility uncertainty needs G-expectation prerequisites and carries difficulty-5 theorems that can fail outright. This topic’s spine uses tools half-known already, and its finance half, which is the schedule risk, is exactly what the re-cut trims.

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